Theory
Cryptology, Assignment-6

Q.1) Ciphertext ⇒ EVIRE

$P = (C - K) \bmod 26$

For All the 26 keys, possible solutions:

Keys	PlainText
0	EVIRE
1	DUHQD
2	CIGPC
3	BSFOB
4	ARENA
5	ZQDMZ
6	YPCLY
7	XOBKX
8	WNAJW
9	VMZIV
10	ULYHU
11	TKXGT
12	SJWFS

keys	PlainText
13	RIVER
14	GHUDQ
15	PGICP
16	OFSBO
17	NERAN
18	MDQZM
19	LCPYL
20	KBOXK
21	JANWJ
22	IZMVI
23	HYLUH
24	GXKTG
25	FWJSF

When k = 4,

E = (4-4) mod 26 = 0 → A
V = (21-4) mod 26 = 17 → R
I = (8-4) mod 26 = 4 → E
R = (17-4) mod 26 = 13 → N
E = (4-4) mod 26 = 0 → A

When k = 13,

E = (4-13) mod 26 = 17 → R
V = (21-13) mod 26 = 8 → I
I = (8-13) mod 26 = 21 → V
R = (17-13) mod 26 = 4 → E
E = (4-13) mod 26 = 17 → R

Plain Text: ARENA PlainText: RIVER.

a) What are the meaningful plaintexts Antony obtains?

Ans: ARENA & RIVER.

b) Where should Antony meet Caesar?

Ans: At the RIVER ⇒ TIBER RIVER

ARENA ⇒ COLOSSEUM.

Both are meaningful, so Antony can't decide from the ciphertext alone.

c) What interesting point about a ciphertext-only attack do you learn from this exercise?

Ans: A ciphertext only attack can give more than one meaningful plaintext when the ciphertext is short.

A longer ciphertext is needed for unique answer.

Q.2) Chosen-Plaintext Attack {Use Affine Cipher (mod 26

HAHAHA → NONONO

a) H = 7

A = 0

N = 13

O = 14

Affine cipher ⇒ $C = (K_1 \cdot P + K_2) \bmod 26$.

$K_1 = a$; $K_2 = b$

b)

$$7a + b \equiv 13 \bmod 26$$
$$0a + b \equiv 14 \bmod 26$$

$$7a \equiv -1 \bmod 26$$

$$a = -(7^{-1}) \bmod 26$$

$$a = -15 \bmod 26$$

$$\Rightarrow \boxed{a = 11}$$

then, $7(11) + b = 13 \mod 26$

$\Rightarrow b = -77 + 13 \mod 26$

$b = -64 \mod 26$

$\Rightarrow \boxed{b = 14}$

q) $\boxed{a = 11}$ and $\boxed{b = 14}$

d) $C = 11(7) + 14 \mod 26$

$\Rightarrow C = 13.$

Q.3) CRWWZ ; encrypted using Affine Cipher: mod 26

Attacker knows that plaintext begins with HA.

a)

PlainText	CipherText
H = 7	C = 2
A = 0	R = 17

WKT, $C = (a.P + b) \mod 26.$

$7a + b = 2 \mod 26$ —①

$0a + b = 17 \mod 26$ —②

$7a = -15 \mod 26$

$a = 15 \ (-15 \mod 26)$

$a = -225 \mod 26$

$\Rightarrow \boxed{a = 9}$

then, $b = 17 \mod 26$

$\boxed{b = 17}$

b) $C = (K_1 P + K_2) \mod 26$

$C = 9 * P + 17 \mod 26.$

c) $P = K_1^{-1} . (C - K_2) \mod 26.$

d) For W (22);

$P = K_1^{-1} \cdot (C - K_2) \bmod 26$

$= 9^{-1} \cdot (22-17) \bmod 26$

$= 3 \cdot (5) \bmod 26 \Rightarrow P = 15 \rightarrow P$

For Z (25);

$P = 9^{-1} \cdot (25-17) \bmod 26$

$= 3 \cdot 8 \bmod 26$

$\Rightarrow P = 24 \rightarrow Y$

Plain Text: — H A P P Y.

Q.4) a) Plain Text:- RIVER – R = 17, I = 8, V = 21, E = 4, R = 17

NQXTP Key:- N = 13, Q = 16, X = 23, T = 19, P = 15

b) Encrypt,

$C_i = (P_i + K_i) \bmod 26$

R + N ⇒ (17 + 13) mod 26 = 4 → E

I + Q ⇒ (8 + 16) mod 26 = 24 → Y

V + X ⇒ (21 + 23) mod 26 = 18 → S

E + T ⇒ (4 + 19) mod 26 = 23 → X

R + P ⇒ (17 + 15) mod 26 = 6 → G

Cipher Text: EYSXG.

c) Decrypt,

$P_i = (C_i - K_i) \bmod 26$

(4-13, 24-16, 18-23, 23-19, 6-15) mod 26

⇒ (17, 8, 21, 4, 17)

Value corresponds to RIVER.

d) Randomness:

A truly random key makes ciphertext unpredictable; patterns in a non-random key can reveal patterns in the message.

Equal: Each plaintext character needs its own key character. A shorter key must be reused (or) repeated, which can create exploitable patterns.

Using Only One: Reusing a key lets attackers compare ciphertexts and potentially learn relationships between the plain texts.